

LAST NAME: _____

Practice Quiz 06D — MATH 2055 — Fall 2026

Practice for Quiz 06

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. Let U be a set and let $A \subseteq U$. The set $U \setminus A$ satisfies the following two conditions on B :

$$A \cap B = \emptyset \quad \text{and} \quad A \cup B = U.$$

Prove that no other subset B of U satisfies both conditions.